

A reusable toolkit that fuses voice +
gesture to manipulate objects in
demo apps

Multimodal Object Manipulation Toolkit

Problem

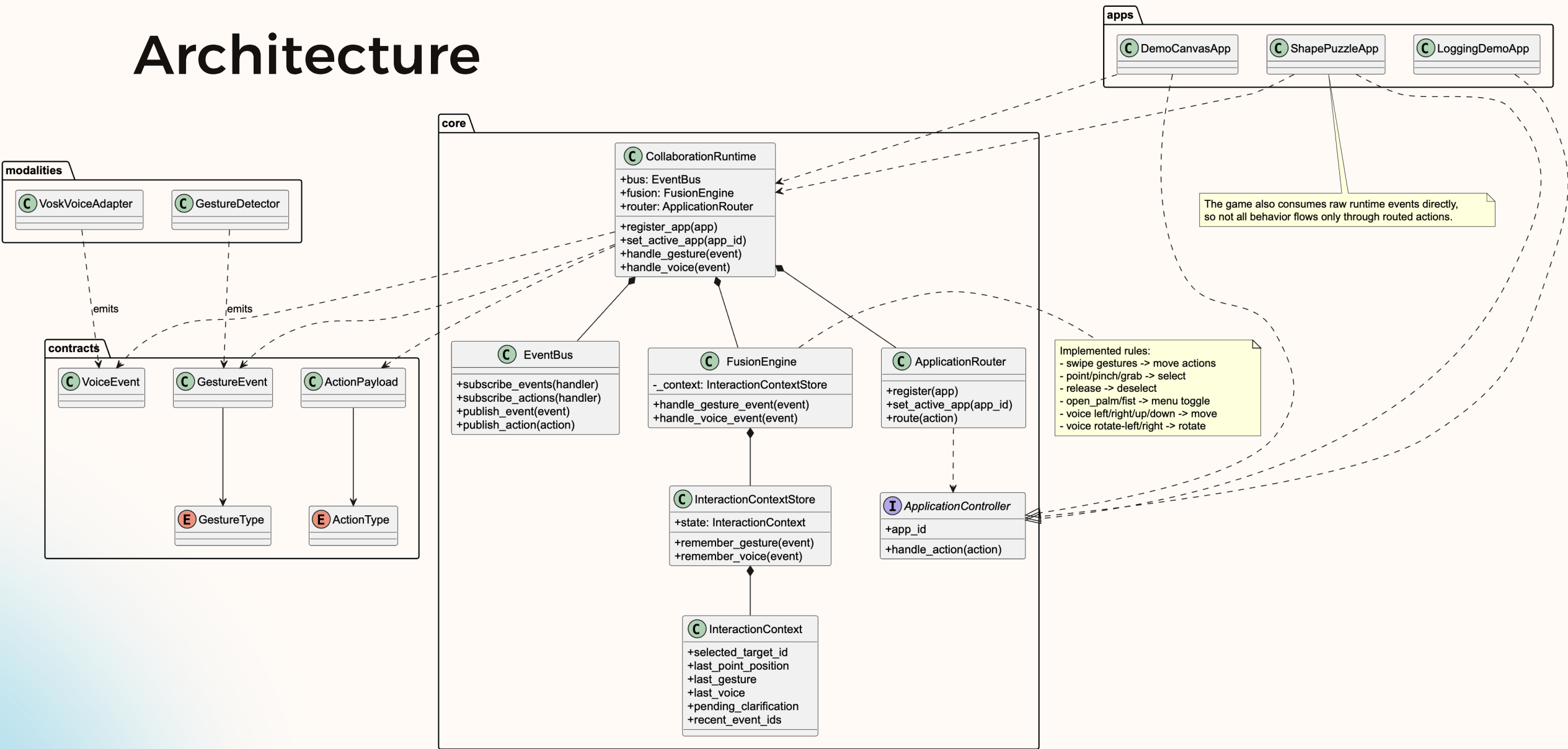
Interacting with 2D objects via single modality is limited/ambiguous

Solution

- Combine modalities to make object manipulation more natural
- Support different intents such as left/right/up/down/rotate-left/rotate-right

Architecture

Core Implementation View



Libraries used for Voice

Microphone access: PyAudio

Provides low level access to the microphone

Text-to-Speech: Vosk

An offline Text-to-Speech library

Microphone Capture + Audio Chunking: SpeechRecognition

Provides an AudioData object from microphone input In this context chunking means capturing a single utterance

Libraries used for Gesture

Mediapipe

Hand-landmark Detection
(21 points per hand).

Opencv-python

Webcam capture and frame
tracking/reading.

Pydantic

Event validation and JSON
Serialization.

Threading

Background gesture detection loop.

Collections.deque

Used for swipe detection by rolling
frame history.

Multimodal 3D Shape Puzzle Game

An interactive puzzle game where users control 3D objects using voice commands and hand gestures, instead of using a keyboard or mouse.

Implementation

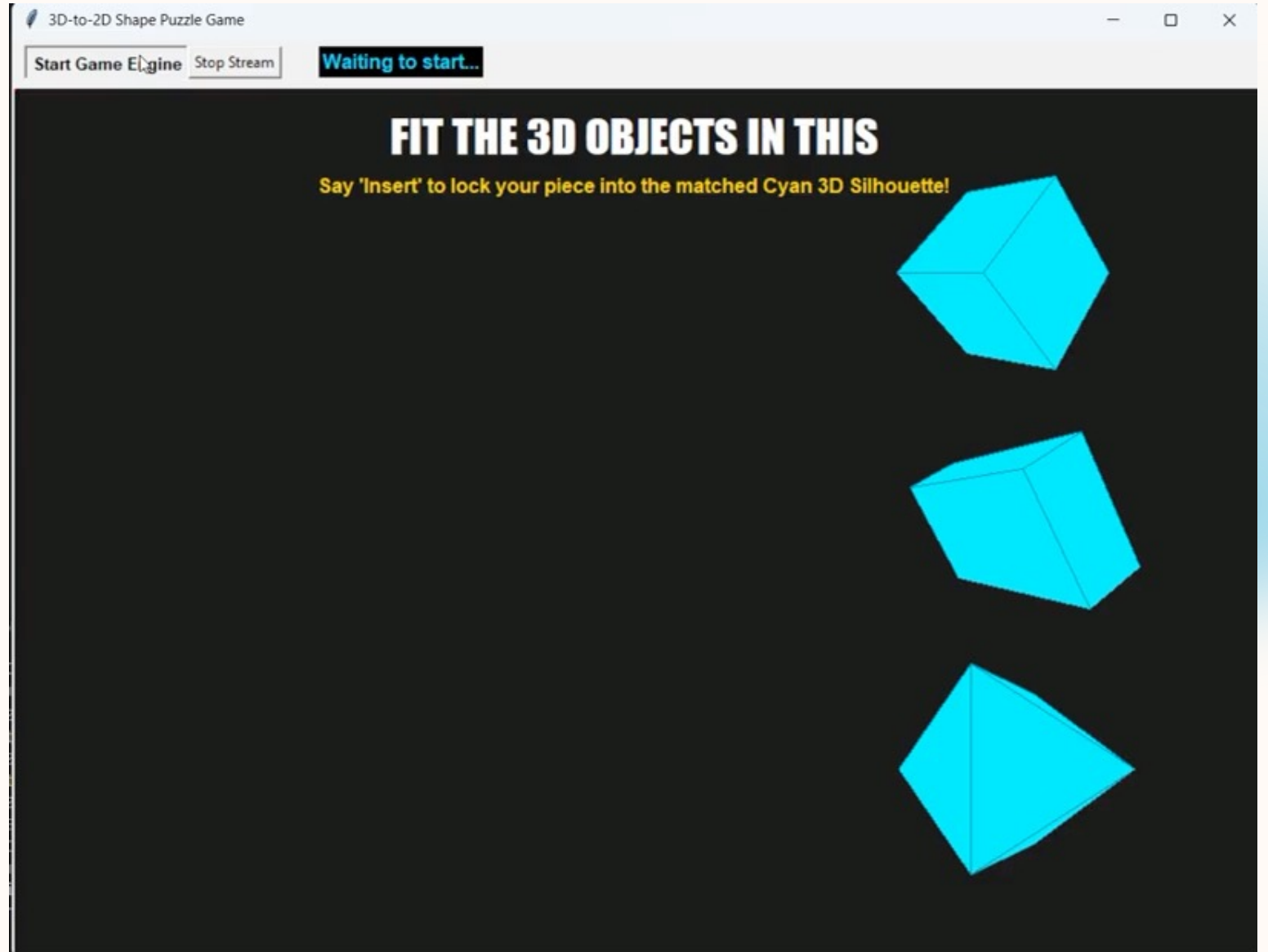
Using Multimodal Toolkit as base.

Additionally used:

1. Tkinter → Used for building the graphical user interface
2. Mathematical Functions / 3D Logic → Used for rotation, scaling, and projection of 3D objects



Demo



Next steps

- 01 Extend engine to fully support all apps.
- 02 Refine weights/arguments/models/fusion engine, so less faulty choice happens. For verification use pre-recorded videos/soundtracks, and pre-recorded final positions and values.
- 03 Make user evaluation for specific use-case (KYC process).
- 04 Create documentation and final presentation.

**Thank
you!**